

Implement histogram equalization on the given image and compare it with the original image using Open CV.

Code:

```
import cv2, urllib.request
from google.colab.patches import cv2_imshow
# Download sample image
urllib.request.urlretrieve(
    "https://raw.githubusercontent.com/opencv/opencv/master/samples/data/lena.jpg",
    "img.jpg"
)
# Read image
img = cv2.imread("img.jpg")
# Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
# Histogram Equalization
equalized = cv2.equalizeHist(gray)
# Display images
print("Original Image")
cv2_imshow(gray)
print("Equalized Image")
cv2_imshow(equalized)
```

Output:

